AI-Based Food Recognition with Nutrition-Aware Recommendations: A System and Reproducibility Audit

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***Abstract*—An image-based food diary must distinguish a detected label, a database nutrient estimate, an assumed portion, and recorded consumption. This paper presents an educational Indian-food application and audits the evidence supporting it. The system combines YOLO11s detection, per-100-g nutrition lookup, user-estimated gram portions, an explicitly saved browser journal, and context-labeled advice. A read-only census of 48,693 exported images across 72 classes identifies 138 exact-duplicate groups spanning the archived partitions and malformed annotation records in 65 images. Source-byte matching and within-export filename-family grouping connect 5,548 candidate groups across those partitions. A review-only reassignment retains class representation while quarantining 11,002 records; it does not establish independent test data for the existing checkpoint. Nutrition lookup is available for 19/72 classes under normalized exact matching, 55/72 without precomputed mappings, and 72/72 with the current mapping layer. These are availability counts, not correctness estimates. The study contributes an inspectable system description, fingerprinted audit artifacts, and a concrete recovery protocol. It does not establish a novel detector, improved logging behavior, measured food mass, or medical efficacy. Independent detector evaluation, qualified mapping review, and asset-permission decisions remain necessary before broader claims.**

***Index Terms*—Food recognition, nutrition databases, data leakage, reproducibility.**

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Short companion to the full review draft; not an external submission or a separate study.

I. Introduction

A food-diary interface must distinguish a model's label, a database nutrient estimate, an assumed portion, and recorded consumption. Correct arithmetic does not ensure that a detected dish matches a database recipe or that the amount entered was eaten. Recognition, mapping, quantity estimation, and advice therefore require different evidence.

We examine the existing *AI-Based Food Recognition with Nutrition-Aware Recommendations* application as an engineering case study. It joins YOLO11s detection, nutrition lookup, gram-based review, explicitly saved browser history, and health-context labels. The detector, data, and inherited dietary rules were preserved during the review.

The paper asks: RQ1, what observable identity and annotation problems occur in the archived export; RQ2, what candidate source grouping establishes; RQ3, how lookup availability changes across three methods; and RQ4, which application invariants are tested. Its contributions are a contract-accurate system description, fingerprinted audit and recovery artifacts, and a measured availability comparison. It does not claim a new detector or a clinically validated recommender.

II. Related Work

Food computing spans recognition, retrieval, recommendation, and monitoring [1]. Im2Calories already integrated image analysis with food-diary and nutritional-estimation tasks [2]; recognition followed by nutrition lookup is not new. IndianFoodNet addresses regional food detection [3], while Indian Thali combines multi-dish recognition with weight and nutrition estimation [4]. Their settings are not matched baselines for this export.

Nutrition5k includes component-weight and nutrition references [5]. Here, grams are entered by users and nutrient values come from a database, so the application has no equivalent nutrient-measurement evaluation. The Indian Nutrient Databank (INDB) supplies recipe-based composition data, not evidence that each photographed preparation matches the chosen row [6]. Its corrigendum corrects source attribution rather than the application's nutrient estimates [7].

Leakage and non-independent evaluation can undermine machine-learning claims [8]. Near-duplicate studies also show why exact equality is narrower than screening transformed or recompressed copies [9]. We use hashes and filename families as auditable diagnostics, not a complete independence certificate. No performance effect from another dataset is assumed here.

A recent FKG.in preprint separates structural, semantic, and source-fidelity validation of LLM-extracted Indian recipes [10]. That knowledge-curation task is relevant context, not a matched benchmark for this application's label-to-row mapping.

III. Implemented System

A. Detection and processed-image alignment

The system uses an existing Ultralytics YOLO11s checkpoint [11], a static Next.js frontend and a FastAPI backend. CPU inference operates at image size 640, confidence 0.25

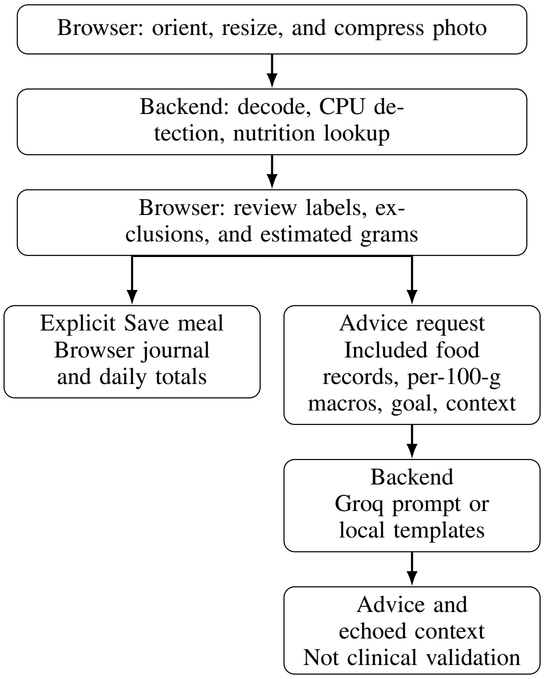

Fig. 1. Separate recording and advice paths. Saving is not triggered by analysis. Selected grams and saved history do not enter the advice request; Groq receives only the name/energy subset of its food records.

and non-maximum-suppression IoU 0.45. Legacy higher class-specific thresholds do not match this checkpoint's lower-snake-case names and therefore do not apply. Confidence is not calibrated nutritional certainty.

The browser normalizes orientation, resizes the longest side to at most 1,600 pixels and JPEG-compresses to at most 3.8 MiB. It displays the same processed image that is uploaded. Backend decoding independently limits supported image formats, bytes to 4 MiB, and decoded pixels to 25 million. The browser's analysis budget is 120 seconds. Input controls and coordinate alignment do not establish detection accuracy.

B. Nutrition, grams, and saved history

SQLite contains 1,010 food rows and 103 mapping entries, including legacy labels; all 72 current classes have precomputed mappings. The local workbook is byte-identical to the original authors' pinned public INDB artifact [12]. Five supplemental records use product/recipe sources for Bhakarwadi, Ghevar, Jalebi, Khandvi, and Nandu Kari [13], [14], [15], [16], [17]. Identity and transcription consistency do not establish recipe equivalence or redistribution permission.

The offline builder prioritizes developer mappings, then normalized exact/alias-containment matching with outer threshold 0.78. Runtime uses precomputed entries, normalized equality, scored containment, and alias-variation containment, followed by the same helper. The helper is not edit-distance or learned semantic matching. One variation stage returns the first containment match without a score gate; set-derived variation order can affect ambiguous cross-process selections. The last helper's internal 0.5 minimum is stricter than its outer 0.4 gate. Current precomputed canonical mappings bypass these fallbacks. Calling the whole pipeline deterministic or clinically conservative would overstate its properties.

Let I_m contain included detections in draft m, let $\mu(c)$ return a row index or $\perp$, and let $K_m = \{q \in I_m : \mu(c_q) \neq \perp\}$. For each database row, let $\mathbf{v}_r$ be the column vector of per-100-g nutrient values. User-estimated grams w_q give

$$\widehat{\mathbf{T}}_m^{known} = \sum_{q \in K_m} \frac{w_q}{100} \mathbf{v}_{\mu(c_q)}, \tag{1}$$

$$u_m = \mathbf{1}\{I_m \setminus K_m \neq \varnothing\}. \tag{2}$$

The sum never indexes an unavailable row. Unknown nutrients mark the known subtotal incomplete rather than being assigned zero. Weights default to 100 g and range from 25 to 1,000 g in 25 g steps. Exclusions remove incorrect detections from calculations; food relabeling is not implemented.

Only explicitly saved meals on the stored local calendar date contribute to daily totals. Saving updates by draft identity, avoiding duplicate-click records; edits preserve the original date. New scans create new drafts. Day selection refreshes at local midnight and on focus/visibility changes. Version-1 browser history retains draft food records including excluded items with their inclusion flags, plus goal preferences, but no photos, health selections, boxes or confidence. It is not a full scientific provenance archive. Corrupt or unsupported storage is not silently overwritten, and scanning remains usable when saving fails.

C. Advice context is not clinical validation

Advice requests contain up to ten included food records, their per-100-g macros, a goal, and session-only health context. The backend's Groq prompt retains names and rounded kcal per 100 g, not numeric protein, carbohydrate or fat. Neither stage receives grams, target calories, consumption totals, or saved history. Separate goal-target calculations normalize policy weights and apply integer rounding; they are not validated prescriptions.

Production uses asynchronous Groq with a 20-second timeout and disabled automatic retries, followed by local templates. Local fallback uses the first two names, with condition templates taking precedence over goal templates; nutrient quantities do not select its advice. Retained template wording can still assume consumption or imply suitability, so fallback and prompt constraints are not a safety guarantee.

Changing context immediately hides old advice, aborts superseded requests, and discards late results. Only a successful matching response supports a badge such as *Adjusted for diabetes*, *Adjusted for high blood pressure*, or *General nutrition guidance*. The echoed context establishes request/response consistency, not expert approval.

IV. Audit Methods

A. Scope and strict checks

Five source exports contain 40,797 locally indexed images: Indian food detection v1, Indian food v6, IndianFood-7 v2, IndianFoodNet v1, and South Indian food detection v19 [18], [19], [20], [21], [22]. Versions and URLs are recovered from local metadata. The old builder pooled source partitions, normalized labels, split sample indices and added further training augmentation. Some sources already contained augmented variants; its 70/15/15 target was not the final export ratio.

The read-only audit records exported image/label SHA-256 hashes, dimensions, accepted class indices, flags, and EXIF-oriented RGB pixel identity without resizing. Cross-partition groups are counted separately for byte and pixel identities. Equality does not rule out other near duplicates.

Each nonblank label row must have five fields, an integer class in range, finite coordinates, centers in $[0, 1]$ and widths/heights in $(0, 1]$. These are syntax/range checks, not semantic annotation validation. They do not verify complete object coverage or box edges inside the frame. Malformed rows without accepted instances are not verified negative examples.

B. Candidate grouping

Recovery rereads exported records against the saved manifest and rejects stale data or class ordering. Source images are indexed by byte hashes and within-export filename prefixes before the earliest `.rf.<32 hex>` marker; source pixels are not separately hashed. Union-find joins equal bytes, equal exported pixel identities, and shared source-family keys, including links through source-only records.

Whole components are quarantined for unresolved lineage, unrecognized source naming, annotation/decode flags, no accepted instances, or equal exported pixels with different label-file hashes. Byte inequality alone cannot distinguish formatting from numeric changes; the follow-up finds none of the direct conflicts to be formatting/order-only. No file is deleted.

Eligible components are assigned greedily to 70/15/15 targets over eligible images, balancing total images and per-class image counts. Seed 42 controls hash-based tie order. Each component remains intact, and component sets in every pair of partitions must be disjoint. This is a checked construction invariant, not proof that filename heuristics found all related photographs. Reassigning data already used in development cannot independently evaluate the current checkpoint.

C. Lookup comparison and software tests

For the fixed 72 labels and database, the audit compares normalized exact matching, automatic runtime lookup with precomputed entries cleared in an isolated instance, and the unchanged current mapping layer. Availability is the fraction returning a row. The database remains read-only; no independent nutrient reference is created.

Backend/contract tests, frontend arithmetic/transport tests and mocked browser tests examine gram arithmetic, exclusions, missingness, explicit saving, edits, deletion/undo, persistence, midnight, storage failures, uploads, errors, response-context switching and stale requests. Desktop and 360-pixel mobile/keyboard paths are covered. Passing these scenarios does not establish usability benefit, clinical quality, or independent detector accuracy.

TABLE I
ARCHIVED EXPORT CENSUS UNDER STRICT AUDIT CHECKS.

Split	Images	Instances	Flagged
Train	36,852	56,287	51
Validation	5,922	8,805	7
Test	5,919	8,868	7

TABLE II
CANDIDATE ASSIGNMENT. EACH ELIGIBLE SPLIT REPRESENTS ALL 72 LABELS; INDEPENDENT ADEQUACY IS UNPROVEN.

Status	Groups	Images
Train	10,251	26,384
Validation	2,203	5,655
Test	2,205	5,652
Quarantine	9,274	11,002

V. RESULTS

A. RQ1: Export integrity

The export contains 48,693 file records, not necessarily unique original photos. The counts in Table I separate image records from accepted annotation instances. Both exact-byte and oriented-pixel checks identify 138 crossing groups, affecting 276 records: 63 train–validation, 56 train–test and 19 validation–test groups. The amount of metric inflation is unknown. Strict flags affect 65 images, including zero-area boxes. The archived evaluator's 8,812/8,875 validation/test instance counts include seven more rows per split than the strict audit; the two counting procedures must not be conflated.

B. RQ2: Review-only reassignment

Source recovery forms 23,933 components containing exports and finds 5,548 candidate lineage groups spanning the old splits. This is heuristic linkage, not visually confirmed identity for every family. Among 37,691 eligible records, 191 remain in excess of distinct pixel identities within their partitions. Quarantine affects 11,002 records by whole-group propagation. Nonexclusive reason counts are 9,127 unresolved-lineage records, 1,455 label-byte conflicts, 542 annotation/decode flags in a group, and 424 a group member without accepted annotations. These are not independently faulty-image counts.

Two earlier full reads returned identical inventory and assignment hashes in the same environment; the second script added input validation. This demonstrates same-environment repeatability, not independent replication. Crops, transformed copies, generic-name collisions and upstream augmentation still require review. No detector was retrained on these candidates.

C. RQ3: Availability is not accuracy

The current layer returns rows for every canonical class, but no independent reference judgments establish which added matches are correct. 21 mappings have lower or missing heuristic scores; higher scores are not expert validation. The

TABLE III
MEASURED ROW AVAILABILITY, NOT CORRECT MAPPING RATE.

Method	Available	Percent
Normalized exact	19/72	26.39
Automatic without precomputed	55/72	76.39
Current mapping	72/72	100.00

counts describe a fixed vocabulary census, not generalization to unseen labels. No supplemental-removal or macro-verification-removal ablation was measured.

D. RQ4: Implemented invariants

Recorded checks passed backend, frontend and mocked browser workflows, including saved-only totals and context matching. Tests establish these software behaviors, not advice quality or reduced logging burden. Health selections and photos are absent from persisted journal records; application code does not deliberately retain uploads, but third-party retention is governed by provider policies.

Inference has one worker and one waiting slot. Provider capacity admits at most two concurrent recommendation operations with no waiting queue. Instance-local rate limits are operational protections, not global spending controls. Readiness fields and release acceptance limits are not benchmark distributions or an availability guarantee.

E. Validation follow-up

Fresh census/grouping runs reproduced all earlier identities and assignments. The 65 zero-area flags comprise 45 zero-width and 20 zero-height rows; all originals were visually inspected with AI assistance, not qualified annotation approval. Numerical comparison of 104 identical-pixel groups with different labels found genuine box/class/instance differences, not only formatting. An additional edge check flags 3,794 rows in 3,346 images for review, not automatic rejection. Minimum class-specific candidate-group support is 43/9/11. A bounded 2,020-record perceptual screen yielded 16 cross-partition candidates for human review; synthetic crop/mirror/rotation probes were missed. These results do not approve a new benchmark.

An additional fingerprinted pass reconciled all 1,010 database rows and 4,040 nutrient fields against the documented workbook import and supplemental literals. Independent rational oracles found no discrepancies in 162,036 goal cases per implementation and 2,880 frontend portion cases at declared tolerances. Eight hash seeds over 181 labels in two lookup modes exposed 30 changing method/label pairs; current canonical mappings stayed stable. The offline advice matrix covered 216 cases, SDK/mock transport covered 16 cases, and extended browser checks covered all 36 goal/context combinations and out-of-order responses. These are implementation/transcription checks, not independent mapping judgments or medical validation. Local Windows execution is distinct from the locked Linux deployment; the full draft records limitations and retained failed harness attempts.

VI. LIMITATIONS AND EVALUATION PLAN

The audit's strongest result is observable non-independence, not an accuracy improvement. Exact grouping can miss transformed relations; filename families can overmerge; broader family-level annotation conflicts remain unresolved. Known group separation is therefore necessary for this candidate protocol but insufficient to certify independent data. Negative-image performance, semantic annotation quality, and the inflation of archived detector scores remain unmeasured.

User-entered grams and named recipe rows do not establish meal composition. Advice sees only limited information, and local templates retain unvalidated wording. Context badges and deterministic arithmetic must not be presented as clinically safe personalization. No user study demonstrates better logging, adherence, or health outcomes.

Before new detector claims, review group links, representative selection, labels and asset rights; freeze a manifest; then retrain appropriately or obtain untouched external evaluation data. Record checkpoint hashes, package versions, operating settings, pretraining exposure and model-selection rules. Comparisons require matched baselines and uncertainty accounting for source-family dependence.

For mapping correctness, freeze outputs and database before qualified, preferably blinded reference review. Define acceptable recipes, approximations, unsupported and ambiguous labels. The new version-2 packet preserves two initial reviews and separate adjudication, while explicitly excluding indeterminate cases from scoring. Report correct/wrong matches, correct abstentions, and missed supported cases rather than availability alone. Forms remain blank; validation of fields and attestations cannot prove expertise or independence. Advice quality and user benefit need separate prespecified assessment and institutional-review consideration when applicable.

VII. CONCLUSION

The application provides inspectable recognition, database estimates, gram review, saved-only history and advice-context handling. Its audit finds 138 exact cross-partition groups and 65 flagged images; candidate lineage reconstruction exposes broader relationships and explicit quarantine. Current lookup covers all 72 labels, without independent correctness evidence. The next milestone is reviewed evaluation and permission/author approval, not additional interface features.

AVAILABILITY AND REVIEW TRANSPARENCY

The full supervisor draft documents methods, runtime caveats and the prospective protocol. The local audited checkout is identified by this repository reference, whose current public accessibility was not verified: source revision 0a4edfa. Full manifests remain local under `.deployment/research/` and must be supplied separately when permissions permit; they are not included in a public checkout or this PDF. The new validation supplement accompanies this copy and is not part of the linked base revision. Source/provider attribution is not blanket permission to redistribute datasets, weights or derived nutrition data.

AI-assisted tools supported inspection, reference checks and editorial revision. Authors must verify the paper and confirm authorship, correspondence, contributions, funding, competing interests and permissions before any external submission. No clinical experiment, participant study or new detector training was conducted in this review.

Appendix A
Historical Scores, Not a Clean Benchmark

The final-training notebook's validation/test outputs report rounded mAP_{50} of 0.830/0.820 and $\mathrm{mAP}_{50:95}$ of 0.692/0.680 on the overlapping old partitions. Its environment is Ultralytics 8.4.60, Python 3.12.13, Torch 2.10.0+cu128 and Tesla T4. Evaluation did not explicitly override confidence or NMS settings, and its roughly 11 ms/image GPU timing is not website CPU latency. The notebook did not record the checkpoint hash at evaluation time. These observations are retained for traceability, not independent generalization. The full review draft identifies the exact notebook and distinguishes older-checkpoint results.

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